Displacement current & the Ampere paradox

Why magnetostatics breaks on a charging capacitor, how Gauss law restores continuity, and the single-surface rule.

Why static Ampere law breaks during charging

Draw one circular loop C around the wire feeding a charging parallel-plate capacitor. Flat surface S1 cuts the wire (current Ic), whereas bulging surface S2 passes between the plates and cuts zero conduction current.

$$\oint_C \mathbf{B}\cdot d\ell = \mu_0(I_c + I_d), \qquad I_d = \varepsilon_0 \tfrac{d\Phi_E}{dt}$$

WHY / One closed rim C cannot have two different circulations. Maxwell-Ampere repairs this by adding changing electric flux through whichever single surface spans C.

WATCH / Never write Ic + Id = 2 Ic on gap surface S2: inside an ideal vacuum gap Ic = 0 and Id = Ic, while on S1 across the wire Id = 0 and Ic = Ic.

Charge conservation & the divergence check

Because div(curl B) is identically zero for every smooth vector field, the magnetostatic law curl B = mu0 J forces div J = 0, contradicting charge accumulation on capacitor plates.

$$\tfrac{\partial \rho}{\partial t} + \nabla\cdot\mathbf{J} = 0, \qquad \nabla\cdot\left(\mathbf{J} + \varepsilon_0 \tfrac{\partial \mathbf{E}}{\partial t}\right) = 0$$

WHY / Substituting Gauss law rho = eps0 div E makes the total effective current J + Jd strictly divergence-free everywhere, so all spanning surfaces give identical flux.

Worked check (Q17): if signed electric flux grows at 4.00e8 V m/s across a fixed vacuum surface, Id = (8.85e-12)(4.00e8) = 3.54 mA.

Capacitor identities: Id = dq/dt = C dV/dt

In the quasistatic parallel-plate model (plate separation d << radius a, variation time tau >> a/c), E = q/(eps0 A) is uniform across area A, so Phi_E = E A = q/eps0.

$$I_d = \varepsilon_0 \tfrac{d(EA)}{dt} = \tfrac{dq}{dt} = I_c = C\tfrac{dV}{dt}$$

WHY / Displacement current has the SI unit ampere and generates the exact same magnetic field as conduction current, even though no electrons cross the vacuum gap.

WATCH / For AC drive V(t) = V0 sin(wt), electric flux and voltage peak at wt = pi/2 while Id = w C V0 cos(wt) peaks at wt = 0: Id leads V by 90 deg.

Steady DC, discharge & sign rules

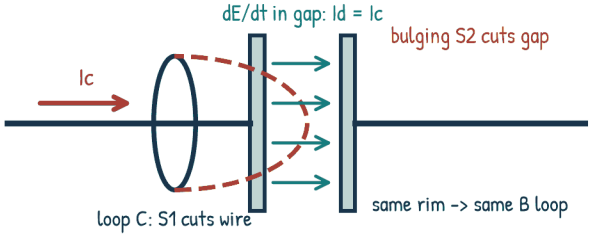
Link loop traversal and surface normal by the right-hand curl rule: if fingers curl along C, thumb sets the positive normal for BOTH conduction and displacement flux.

$$\text{Steady DC}: \ \tfrac{dV}{dt} = 0 \Rightarrow I_d = 0; \qquad \text{Discharge}: \ I_d = \dot{q} < 0$$

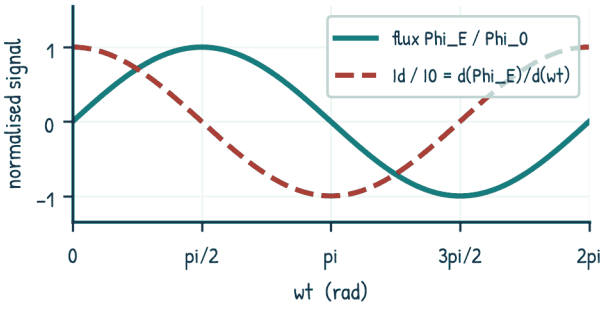
WHY / During capacitor discharge through a resistor, E decreases while E still points from + to - plate, so dPhi_E/dt < 0 reverses B alongside reversing wire current.

WATCH / For moving plates at constant charge q, E = q/(eps0 A) stays fixed so Id = 0, even though plate voltage V = E d(t) is changing!

One Amperian rim C spanned by flat surface S1 (Ic) or gap surface S2 (Id)



Sinusoidal plate voltage: displacement current leads electric flux by pi/2



REMEMBER / Evaluate Ic and Id across the SAME spanning surface of loop C; adding both from two surfaces double-counts one current.

Capacitor gap magnetic field & dielectrics

Radial profile of B_phi inside and outside circular plates, rim continuity check, and bound polarisation current in matter.

Azimuthal B_phi inside and outside circular plates

For circular plates of radius a carrying charging current Ic, a coaxial circle of radius r < a encloses fraction (pi r^2)/(pi a^2) = r^2/a^2 of the uniform gap flux.

$$B_\phi(r) = \frac{\mu_0 I_c r}{2\pi a^2}\ (r \le a), \qquad B_\phi(r) = \frac{\mu_0 I_c}{2\pi r}\ (r \ge a)$$

WHY / Plate spacing d cancels completely! On the symmetry axis r = 0 so B = 0; at rim r = a both formulas match cleanly at B_max = mu0 Ic / (2 pi a).

Worked exemplar (E1): a = 0.050 m, Ic = 0.020 A gives B(0.02 m) = 32 nT, rim peak B(0.05 m) = 80 nT, and outside B(0.10 m) = 40 nT.

Macroscopic displacement current in a dielectric

Fill the capacitor uniformly with linear nonmagnetic dielectric eps = kappa eps0. Free plate charge q_f sets D = q_f / A = eps0 E + P_pol across the gap.

$$I_c = A\frac{dD}{dt} = \varepsilon_0 A\dot{E} \,(I_c/\kappa) + A\dot{P}_{\text{pol}}\,((1-1/\kappa)I_c)$$

WHY / Bound charges shift locally inside molecules, creating genuine polarisation current A dP_pol/dt without free electrons crossing the insulator.

Worked split (E2): kappa = 5 at Ic = 15 mA gives total A dD/dt = 15 mA, split into 3 mA vacuum term + 12 mA polarisation current (sum = 15 mA).

Fixed-Ic vs fixed-(dV/dt) constraint trap

Always check which variable the external circuit holds constant when comparing an empty capacitor to one filled with dielectric constant kappa (C2, Q2):

$$\text{Fixed } I_c:\ B_\phi \text{ unchanged}; \quad \text{Fixed } \dot{V}:\ I_d = \kappa C_0 \dot{V} \Rightarrow B_\phi \propto \kappa$$

WHY / At fixed current source Ic, dD/dt = Ic/A does not change so B_phi is untouched (dE/dt drops by kappa). At fixed voltage ramp dV/dt, current scales by kappa.

WATCH / Never write eps0 A dE/dt = Ic inside a kappa > 1 dielectric: eps0 A dE/dt is only Ic / kappa, missing fraction (1 - 1/kappa) carried by bound charges (Q9).

Leaky resistor-capacitor & validity limits

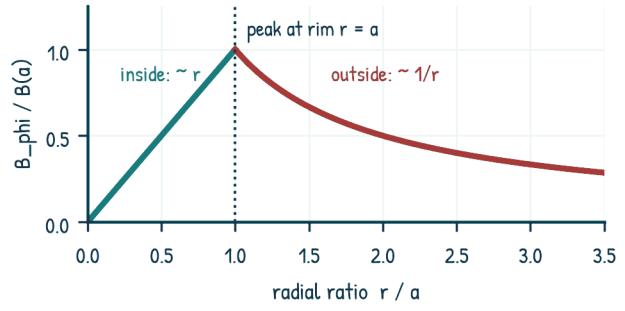
If an Ohmic medium of conductivity sigma and permittivity eps fills the gap under voltage V(t), free conduction current J_f = sigma E and displacement J_d = eps dE/dt coexist:

$$\frac{|J_d|}{|J_f|} = \frac{\omega\varepsilon}{\sigma} = \omega RC$$

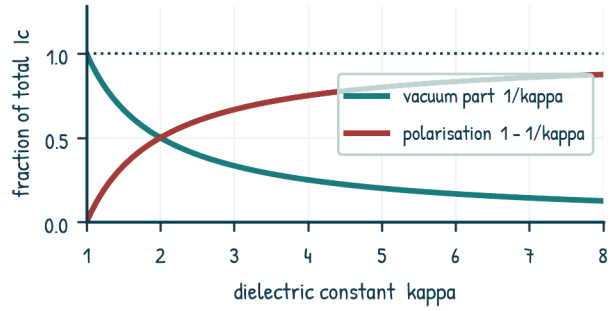
WHY / At low frequencies (w << sigma/eps) conduction dominates; above relaxation frequency w_c = sigma/eps, displacement current wins.

WATCH / Quasistatic formula B_phi = mu0 Ic r/(2 pi a^2) assumes w a << c; at microwave frequencies radial wave variations replace uniform E(t) by Bessel modes.

Gap magnetic field rises linearly inside the plate radius and falls as 1/r outside



Vacuum displacement vs bound polarisation current share in a filled capacitor



REMEMBER / Inside circular plates B_phi rises linearly with r and peaks at rim r = a; in a dielectric, A dD/dt = Ic splits into vacuum + bound parts.

All four Maxwell equations in vacuum & matter

Integral and differential forms side by side, physical meaning of each law, and linear constitutive relations.

Microscopic (vacuum) Maxwell equations

Every classical electromagnetic phenomenon in vacuum is governed by four coupled field laws linking total charge density rho and total current density J:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}, \qquad \nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \qquad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

WHY / Gauss theorem turns div laws into closed-surface fluxes; Stokes theorem turns curl laws into line integrals around fixed loop boundaries.

Macroscopic laws in matter: separate free sources

Absorbing bound charge into D = eps0 E + P and bound currents into H = B/mu0 - M isolates controllable free charge rho_f and free current J_f:

$$\oint \mathbf{D} \cdot d\mathbf{A} = Q_f, \qquad \oint \mathbf{B} \cdot d\mathbf{A} = 0$$

$$\oint \mathbf{E} \cdot d\ell = -\frac{d\Phi_B}{dt}, \qquad \oint \mathbf{H} \cdot d\ell = I_f + \frac{d\Phi_D}{dt}$$

WATCH / Notice asymmetry: magnetic Gauss law keeps B (never H, since div H = -div M != 0 at pole faces), while Faraday keeps E and B unmodified.

Model L constitutive closure & SI constants

A homogeneous, isotropic, linear, lossless, nondispersive medium (Model L) closes Maxwell equations with positive constants eps = eps_r eps0 and mu = mu_r mu0:

$$\mathbf{D} = \varepsilon \mathbf{E}, \quad \mathbf{B} = \mu \mathbf{H}, \quad c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \approx 3.00 \times 10^8 \ \mathrm{m/s}$$

WHY / Check SI constants: eps0 = 8.85e-12 F/m, mu0 = 4 pi e-7 H/m give Z0 = sqrt(mu0/eps0) = 376.7 Ohm ~ 377 Ohm and 1/sqrt(mu0 eps0) = 2.998e8 m/s.

WATCH / Do not plug DC dielectric constant kappa_static (e.g. water ~ 80) into optical index n ~ 1.33: heavy dipoles freeze out at optical frequencies!

Symmetry, causality & moving-loop warning

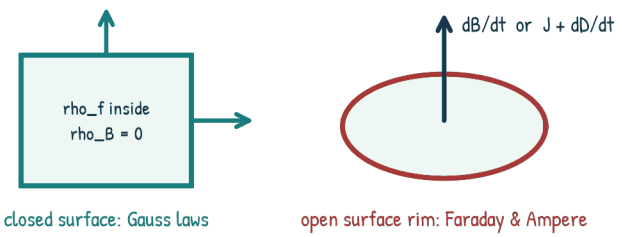
In a travelling wave, "changing E creates B and changing B creates E" is only a verbal mnemonic — not a time lag! Both curls hold simultaneously at each (r, t).

$$\mathcal{E}_{\text{loop}} = \oint (\mathbf{E} + \mathbf{u} \times \mathbf{B}) \cdot d\ell = -\frac{d\Phi_B}{dt}$$

WHY / In source-free space (rho_f = 0, J_f = 0), E and B become symmetric up to duality E -> v B, B -> -E/v, unlocking self-sustaining wave propagation.

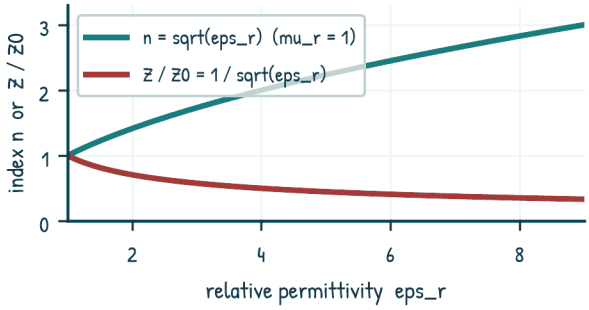
WATCH / Fixed-loop integral curl E = -dB/dt gives only transformer EMF; if wire segments move at velocity u through B, include u x B motional EMF separately.

Gauss laws integrate over closed volumes; curl laws integrate around open rims



closed surface: Gauss laws — open surface rim: Faraday & Ampere

Nonmagnetic dielectrics raise refractive index n while lowering wave impedance Z



REMEMBER / Maxwell unites two divergence laws (sources) with two coupled curl laws (time-varying fields); no magnetic monopoles exist (div B = 0).

Curl-curl derivations, wave speed & refraction

Deriving both vector wave equations, why charge-free homogeneity is essential, and wavelength vs frequency in matter.

Deriving both 3D vector wave equations

In homogeneous source-free Model L (rho_f = 0, J_f = 0, constant eps and mu), take the curl of Faraday law and substitute curl B = mu eps dE/dt:

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\mu\varepsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

$$\nabla^2 \mathbf{E} = \mu\varepsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}, \qquad \nabla^2 \mathbf{B} = \mu\varepsilon \frac{\partial^2 \mathbf{B}}{\partial t^2}$$

WHY / Charge-free constant eps forces div E = 0, killing grad(div E); div B = 0 kills grad(div B) identically in the parallel curl-of-Ampere derivation.

Phase speed & refractive index in matter

Substitute any plane waveform F(z - v t): d^2F/dz^2 = F'' and d^2F/dt^2 = v^2 F'', forcing mu eps v^2 = 1. Compare with vacuum c = 1/sqrt(mu0 eps0):

$$v = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{c}{n}, \qquad n = \sqrt{\mu_r \varepsilon_r}$$

Quick checks: if eps_r = 4 and mu_r = 9 (Q3), n = sqrt(36) = 6 so v = c/6 (never omit mu_r or the square root!). Nonmagnetic matter has mu_r = 1 so n = sqrt(eps_r).

WHY / Dimensional check: [mu0 eps0] = (N A^-2)(C^2 N^-1 m^-2) = C^2/(A^2 m^2) = s^2/m^2, so [1/sqrt(mu0 eps0)] = m/s automatically!

Refraction: frequency stays fixed, wavelength shrinks

Boundary fields at a stationary interface must match at EVERY instant t, forcing exp(-i w t) to carry identical angular frequency w in both media (E3, Q19):

$$f_{\text{med}} = f_0, \qquad \lambda_{\text{med}} = \frac{v}{f_0} = \frac{\lambda_0}{n}, \qquad k_{\text{med}} = n k_0$$

Worked exemplar (E3): f = 6.00e14 Hz (lambda_0 = 500 nm green) enters nonmagnetic glass eps_r = 2.25 (n = 1.50): v = 2.00e8 m/s, lambda = 333 nm, colour stays 600 THz!

WATCH / Never classify a wave inside glass with lambda = 333 nm as "ultraviolet": spectral bands and photon energy E = h f depend on f (or vacuum lambda_0), never lambda_med.

Hidden coupling & inhomogeneous-medium trap

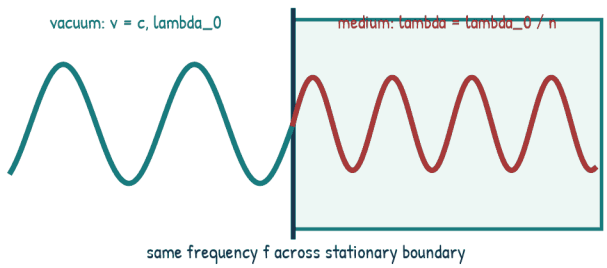
Six scalar components (Ex, Ey, Ez, Bx, By, Bz) each satisfy the wave equation, yet you CANNOT pick them independently (Q10):

$$\nabla \cdot (\varepsilon \mathbf{E}) = 0 \Rightarrow \nabla \cdot \mathbf{E} = -\frac{\mathbf{E} \cdot \nabla \varepsilon}{\varepsilon} \neq 0$$

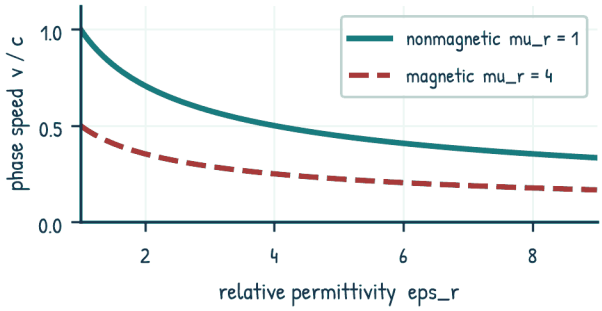
WHY / In graded-index fibres (eps = eps(r)), div E != 0 restores the extra grad(div E) term so simple decoupled wave equations fail (C3).

WATCH / Even in vacuum, Ex = E0 sin(kx - wt) is ILLEGAL because div E = dEx/dx = k E0 cos(kx - wt) != 0 violates Gauss law in charge-free space!

Entering a medium of index n compresses wavelength while preserving frequency f



Phase speed v = c / sqrt(mu_r eps_r): both constitutive factors enter equally



REMEMBER / Wave speed is v = c / sqrt(mu_r eps_r); stationary boundaries keep frequency f strictly constant while scaling lambda = lambda_0 / n.

Plane-wave kinematics, phase & right-handed triad

Reading k and w with correct signs, reconstructing B from E without sign mistakes, and testing valid Maxwell fields.

Reading propagation vector k & speed from phase

Set phase phi(r, t) = k . r - w t = constant (or w t - k . r = constant): differentiating gives k . dr/dt = w, so crests move along +k_hat at speed v = w/|k|.

$$\phi = k_x x + k_y y + k_z z \mp \omega t \Rightarrow \hat{\mathbf{k}} = \pm \frac{(k_x, k_y, k_z)}{\sqrt{k_x^2 + k_y^2 + k_z^2}}$$

WHY / Opposite signs on space and time (kz - wt or wt - kz) mean travel in +z; SAME signs (kz + wt) mean travel in -z! Always check sign first.

Standard links: k = 2 pi / lambda, w = 2 pi f, T = 1/f, and v = w / k = f lambda = c / sqrt(mu_r eps_r).

Transversality & exact vector link B = (k_hat x E)/v

Apply nabla -> i k to all four source-free Maxwell equations for a single monochromatic plane wave in homogeneous Model L:

$$\mathbf{k}\cdot\mathbf{E} = 0, \quad \mathbf{k}\cdot\mathbf{B} = 0, \quad \mathbf{B} = \tfrac{1}{v}(\hat{\mathbf{k}} \times \mathbf{E}), \quad E_0 = vB_0$$

WHY / (E_hat, B_hat, k_hat) form a right-handed orthonormal triad at every instant: E x B always points along +k_hat because real factor 1/v > 0 keeps E and B in phase!

WATCH / In a dielectric of index n > 1, E0 = v B0 = (c/n) B0 (Q11, Q30): blindly writing B0 = E0 / c underestimates magnetic amplitude by factor n!

Worked vector reconstructions without guesswork

Exemplar E4: in vacuum E = 60 cos(2z - 6.00e8 t) x_hat (SI). Here k = +2 z_hat, w = 6.00e8 rad/s -> v = w/k = 3.00e8 m/s = c along +z_hat.

$$\mathbf{B} = \frac{\hat{\mathbf{z}} \times \hat{\mathbf{x}}}{c} E = (2.00 \times 10^{-7}\ \mathrm{T}) \cos(2z - 6.0 \times 10^{8} t)\, \hat{\mathbf{y}}$$

Cross-product drill (C4, Q4): if wave moves along -x_hat and E points along +z_hat, B_hat = (-x_hat) x (+z_hat) = +y_hat. Verify: E_hat x B_hat = z x y = -x_hat!

WHY / Half a cycle later both E and B flip sign simultaneously (-E) x (-B) = +(E x B), so energy continues flowing along -x_hat without reversal.

Four-step validity test for candidate exam fields

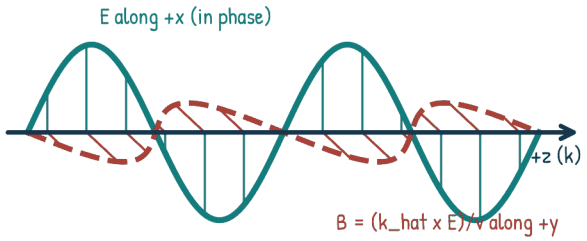
JEE Advanced frequently offers 4 vector expressions and asks which represents a valid plane EM wave in charge-free space. Test in exact order:

$$(1)\ \mathbf{k}\cdot\mathbf{E}_0 = 0, \quad (2)\ \mathbf{k}\cdot\mathbf{B}_0 = 0, \quad (3)\ \mathbf{B}_0 = \tfrac{\hat{\mathbf{k}} \times \mathbf{E}_0}{v}, \quad (4)\ \tfrac{\omega}{k} = v$$

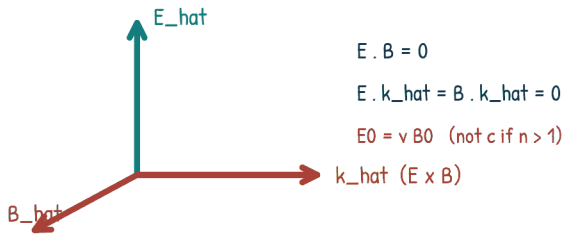
WHY / Any failure in steps 1-4 disqualifies the candidate immediately. E.g. E = E0 sin(kx - wt) (x_hat + y_hat) has k = k x_hat, so k . E0 = k E0 != 0 -> INVALID!

WATCH / For tilted wave vector k = 3 x_hat + 4 y_hat, orthogonality means 3 Ex + 4 Ey = 0, so E0 proportional to (4 x_hat - 3 y_hat) IS valid!

Single travelling plane wave: E and B oscillate in phase and cross into +k



Right-handed triad: reversing BOTH E and B half a cycle later keeps S forward



REMEMBER / Read direction k_hat from constant phase first, verify k . E = 0, then compute B = (k_hat x E) / v; never use c when v = c / n.

Polarisation states, unpolarised light & Malus law

Superposing orthogonal transverse modes, linear/circular/elliptical criteria, and polaroid analyser chains.

Linear, circular & elliptical polarisation criteria

Superpose two orthogonal transverse plane waves propagating along +z with phase difference delta (where phi = kz - wt):

$$\mathbf{E} = E_{0x}\cos\phi\,\hat{\mathbf{x}} + E_{0y}\cos(\phi+\delta)\,\hat{\mathbf{y}}, \qquad E_z = B_z = 0$$

Case 1: delta = m pi -> straight line of slope (-1)^m E_0y/E_0x (linear). Case 2: E_0x = E_0y and delta = +-pi/2 -> circle Ex^2 + Ey^2 = E0^2. All others -> ellipse.

WHY / Even for circular/elliptical waves, instantaneous B = (z_hat x E)/v still has |B| = |E|/v and stays perpendicular to E at every instant!

Unpolarised light & Malus cos^2(theta) law

Natural thermal/lamp light consists of rapid random jumps in phase delta, sweeping all transverse directions equally: <cos^2(theta)> = 1/2.

$$I_{\text{after 1st}} = \tfrac{1}{2}I_0, \qquad I_{\text{after next}} = I_{\text{in}}\cos^2\theta$$

WHY / An ideal linear polaroid projects electric amplitude onto pass axis p_hat: E_trans = E_in cos(theta), and squaring field amplitude gives cos^2(theta) in intensity.

WATCH / Measure theta between consecutive polaroid pass axes (or between incoming E direction and pass axis) — if a problem states angle to the block axis, use sin^2!

Crossed polaroids + rotating middle sheet

Place two crossed polaroids P1 (0 deg) and P3 (90 deg), then insert middle polaroid P2 at angle theta to P1 for incident unpolarised intensity I0:

$$I_{\text{out}} = \tfrac{I_0}{2}\cos^2\theta\cos^2(90^\circ-\theta) = \tfrac{I_0}{8}\sin^2(2\theta)$$

WHY / At theta = 45 deg, I_out peaks at I0 / 8 (non-zero even though P1 and P3 are orthogonal!) because P2 projects a fresh component along 45 deg.

Rotation check: in one full 2 pi revolution of P2, sin^2(2 theta) hits 4 maxima (at 45, 135, 225, 315 deg) and 4 zeros (0, 90, 180, 270 deg).

Circular-wave intensity & near-field warning

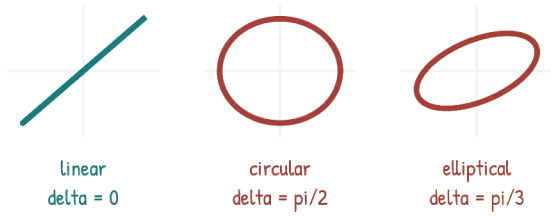
Compare a linearly polarised wave E = E0 cos(phi) x_hat against a circularly polarised wave E = E_c (cos(phi) x_hat + sin(phi) y_hat) of constant magnitude E_c:

$$I_{\text{lin}} = \frac{E_0^2}{2Z}, \qquad I_{\text{circ}} = \frac{E_{0x}^2+E_{0y}^2}{2Z} = \frac{E_c^2}{Z}$$

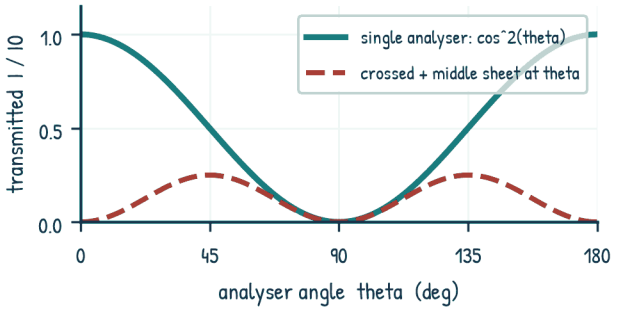
WATCH / In circular polarisation |E(t)| = E_c is constant in time (never drops to zero!), so mean energy density is eps E_c^2 without the 1/2 factor!

WATCH / Longitudinal E components DO exist inside waveguided TM modes, plasma oscillations, and electrostatic near fields — just never in a vacuum plane wave.

Tip trajectory of E in the transverse xy-plane for three phase differences



Malus law cos^2(theta) vs inserting a third polariser between crossed sheets



REMEMBER / Polarisation tracks the transverse E vector; unpolarised light transmits I0/2 through the first ideal polariser and I_prev cos^2(theta) thereafter.

Field energy density & Poynting theorem

Stored electric/magnetic energy densities, exact curl derivation of energy flux S = E x H, and travelling-wave equipartition.

Electric & magnetic energy densities

From parallel-plate storage U = (1/2) C V^2 / (A d) and solenoid storage U = (1/2) L I^2 / (A l), linear nondispersive media store local field energy density:

$$u_E = \tfrac{1}{2}\varepsilon E^2, \qquad u_B = \frac{B^2}{2\mu}, \qquad u = u_E + u_B$$

WHY / Both terms have SI dimensions J/m^3 = N/m^2 = Pa (identical dimensions to pressure!). Total field energy in volume V is integral of u dV.

WATCH / Do not apply u = (1/2) eps E^2 blindly inside ferromagnetic hysteresis loops or strongly dispersive media near resonances where d(w eps)/dw enters.

Poynting vector & exact local conservation law

Apply vector identity div(E x H) = H . (curl E) - E . (curl H) and substitute Faraday + Maxwell-Ampere laws for constant real eps, mu:

$$\mathbf{S} = \mathbf{E}\times\mathbf{H} = \frac{\mathbf{E}\times\mathbf{B}}{\mu}, \qquad \frac{\partial u}{\partial t} + \nabla\cdot\mathbf{S} = -\mathbf{J}_f\cdot\mathbf{E}$$

WHY / Integrating over fixed volume V says: rate of loss of stored EM energy (-dU/dt) equals outward power flux oint S . dA PLUS mechanical/Ohmic power J_f . E on charges.

Units check: [E][H] = (V/m)(A/m) = W/m^2 (power per unit area normal to S). In a DC wire carrying I across voltage V, radial S feeds I^2 R heat from outside!

Instantaneous equipartition in a single plane wave

Substitute B = E / v and v^-2 = mu eps into uB for one travelling plane wave in homogeneous source-free Model L (Q11):

$$u_B = \frac{(E/v)^2}{2\mu} = \frac{\mu\varepsilon E^2}{2\mu} = \tfrac{1}{2}\varepsilon E^2 = u_E \Rightarrow u = \varepsilon E^2 = \frac{B^2}{\mu}$$

WHY / Unlike an LC circuit or standing string where kinetic/magnetic peak when potential/electric is zero, a travelling wave carries uE and uB IN PHASE at every (r, t)!

WATCH / Equipartition uE = uB holds for ONE travelling wave, NOT for static fields, near fields, or standing waves (where E nodes sit at B antinodes!).

Transport velocity link: S = u v k_hat

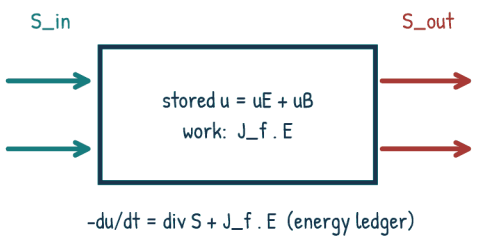
Combine |S| = E B / mu = E^2 / (mu v) with total density u = eps E^2 and mu eps = 1/v^2 for any single plane wave waveform f(kz - wt):

$$\mathbf{S}(z,t) = u(z,t)\, v\, \hat{\mathbf{k}}$$

WHY / In time dt, a cylinder of base area A and length v dt sweeps all energy inside volume A (v dt) across surface A: dU = u A v dt -> flux S = dU/(A dt) = u v.

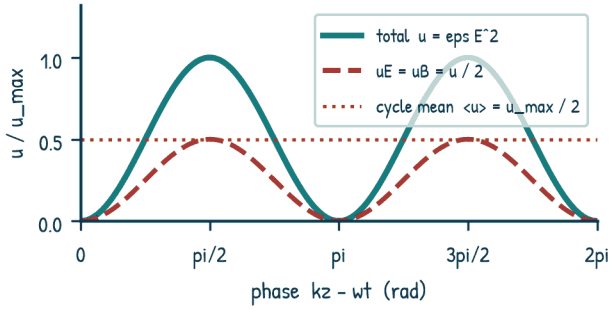
Averaging over one period preserves the exact transport identity: mean intensity I = <S> = v <u> (in vacuum: I = c <u>, so <u> = I / c).

Poynting balance: field energy loss equals outward Poynting flux plus Ohmic work



-du/dt = div S + J_f . E (energy ledger)

Travelling wave: uE and uB peak and vanish together, never trading back and forth



REMEMBER / In a single travelling plane wave uE = uB at EVERY instant (both peak and vanish together), giving S = (E x B)/mu = u v k_hat.

Peak vs RMS amplitude, impedance & 1/r^2 scaling

All six equivalent intensity expressions, vacuum impedance Z0 = 377 Ohm, and point-source vs collimated beam scaling.

All equivalent intensity formulas in one chain

Define wave impedance Z = E/H = mu v = sqrt(mu/eps) = Z0 sqrt(mu_r/eps_r). Since <sin^2(phi)> = 1/2, E_rms = E0/sqrt(2) and B_rms = B0/sqrt(2):

$$I = \langle S \rangle = \tfrac{1}{2}\varepsilon v E_0^2 = \frac{E_0 B_0}{2\mu} = \frac{v B_0^2}{2\mu} = \frac{E_0^2}{2Z} = \frac{E_{\text{rms}}^2}{Z} = v\langle u \rangle$$

WHY / Notice how Z unifies electricity and optics: I = E_rms^2 / Z mirrors resistor power V_rms^2 / R with Z0 = 377 Ohm in vacuum (Z = Z0 / n if mu_r = 1).

WATCH / Never average before multiplying (C5, Q12): <E> = 0 and <B> = 0, yet <E x B> = (E0 B0 / 2) k_hat > 0 because E and B reverse together every half-period!

Worked peak vs RMS & energy-density drills

Exemplar E5: vacuum linear wave with peak E0 = 120 V/m has I = 0.5 * (8.85e-12)(3.0e8)(120)^2 = 19.1 W/m^2, <u> = I/c = 63.7 nJ/m^3, <uE> = 31.9 nJ/m^3.

$$\langle u \rangle = \varepsilon E_{\text{rms}}^2 = \tfrac{1}{2}\varepsilon E_0^2, \qquad \langle u_E \rangle = \langle u_B \rangle = \tfrac{1}{2}\varepsilon E_{\text{rms}}^2 = \tfrac{1}{4}\varepsilon E_0^2$$

Paper checks (Q5, Q21, Q22): E_rms = 30.0 V/m in vacuum gives I = 30^2 / 377 = 2.39 W/m^2 (NO extra 1/2!). Beam I = 240 W/m^2 gives <u> = 240 / 3e8 = 0.800 uJ/m^3.

WATCH / #1 exam error: inserting an extra 1/2 into E_rms^2 / Z0 or forgetting 1/2 in eps0 c E0^2 / 2! Always circle whether question data gives PEAK or RMS!

Point source vs line source vs collimated laser

Energy conservation across enclosing surfaces in lossless space fixes how intensity I and peak amplitude E0 scale with distance r:

$$\text{Point}: \; I = \frac{\mathcal{P}}{4\pi r^2} \; (E_0 \propto r^{-1}); \quad \text{Line}: \; I = \frac{\mathcal{P}}{2\pi r L} \; (E_0 \propto r^{-1/2})$$

Worked beacon (Q32): P = 120 W isotropic at r = 3.00 m gives I = 120/(36 pi) = 1.061 W/m^2, E0 = sqrt(2 I Z0) = 28.3 V/m, B0 = E0/c = 94.2 nT.

WHY / For an ideal parallel laser beam of cross-section pi w^2, I = P/(pi w^2) and E0 remain independent of distance r (neglecting diffraction).

Lamp efficiency & photon-flux bridge

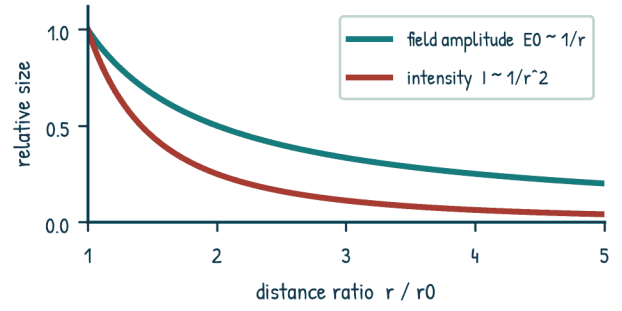
If a bulb of electrical input P_elec converts fraction eta to monochromatic radiation of wavelength lambda_0 isotropically, photon flux phi_ph (photons / m^2 s) at distance r is:

$$I(r) = \frac{\eta \mathcal{P}_{\text{elec}}}{4\pi r^2}, \qquad \Phi_{\text{photon}} = \frac{I(r)}{hc/\lambda_0} = \frac{\eta \mathcal{P}_{\text{elec}} \lambda_0}{4\pi r^2 hc}$$

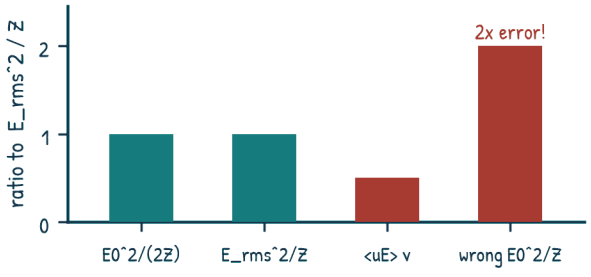
WHY / Links classical Maxwell wave amplitude E0 = sqrt(2 I Z0) directly to quantum counting rate N_dot = Phi_photon A_det cos(theta) on a detector.

WATCH / If light reflects off a plane mirror behind the point source, add the image source at image distance r_img (coherent phase if laser, intensities if thermal)!

Isotropic far field: doubling r halves E0 and B0, while intensity quarters



Peak vs RMS bookkeeping: omit the 1/2 only when squaring E_rms, never E0



REMEMBER / I = E0^2/(2Z) = E_rms^2/Z = v <u>; for an isotropic point source I ~ 1/r^2 so E0 and B0 drop as 1/r, whereas a collimated laser keeps I constant.

Field momentum & flat-surface radiation pressure

Maxwell momentum density g = S/c^2, why cos^2(theta) appears for tilted plates, tangential shear force, and partial reflection.

Field momentum density & pulse impulse

Combining Lorentz force density rho E + J x B with Maxwell stress tensor T_ij in vacuum identifies electromagnetic momentum density g and collimated pulse momentum p:

$$\mathbf{g} = \varepsilon_0(\mathbf{E}\times\mathbf{B}) = \frac{\mathbf{S}}{c^2}, \qquad \mathbf{p}_{\text{pulse}} = \frac{U}{c}\hat{\mathbf{k}}$$

Pulse impulse (E6, Q23): a collimated 9.00 mJ pulse hitting a flat mirror normally reverses momentum +U/c -> -U/c, delivering J = 2 U/c = 6.00e-11 N s (absorber: 3.00e-11 N s).

WATCH / Isotropic or counter-propagating equal beams have total energy U > 0 yet net vector momentum p = 0 (Q12): p = U/c holds ONLY for unidirectional vacuum radiation!

Flat oblique surface: why TWO cosines appear

Let uniform vacuum beam of intensity I strike flat area A_s at angle theta to surface normal n_hat. Energy hitting patch in dt is dU = I (A_s cos theta) dt:

$$\mathcal{P}_{\text{hit}} = IA_s\cos\theta, \qquad P_n = (1+R)\tfrac{I}{c}\cos^2\theta \quad (T=0)$$

WHY / First cos(theta) shrinks intercepted beam cross-section; second cos(theta) projects normal momentum (1 + R)(dU/c)cos(theta). Dividing by actual area A_s gives cos^2(theta)!

WATCH / CRITICAL TRAP (C6, Q25): if a question ALREADY gives intercepted power P_hit (e.g. 12 W beam all landing on a 60 deg mirror), area projection is DONE: F_n = 2(P_hit/c)cos(theta) = 40 nN!

Tangential shear traction & transparent sheets

Decompose incident ray unit vector k_hat = cos(theta) n_hat + sin(theta) t_hat (inward normal n_hat, surface tangent t_hat). Specular rays keep tangential momentum unchanged, absorbed rays transfer it all (Q13, Q33):

$$F_n = (1+R-T)\tfrac{IA_s}{c}\cos^2\theta, \qquad F_t = \mathcal{A}\tfrac{IA_s}{c}\sin\theta\cos\theta$$

WHY / For a pure absorber (A = 1, R = T = 0), sqrt(F_n^2 + F_t^2) = I A_s cos(theta)/c points straight along incoming beam k_hat, NOT normal to the plate!

Worked sail (E7): A_s = 0.020 m^2, R = 0.60, T = 0, I = 900 W/m^2 at 60 deg -> P_n = 1.60 * 900 * 0.25 / 3e8 = 1.20 uPa and F_n = 24.0 nN.

Emissive recoil & laser levitation balance

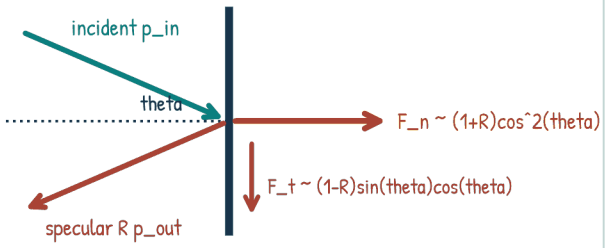
A horizontal mirror disc of mass m levitates atop a vertical upward laser beam of total power P_beam (all hitting disc, normal incidence R = 1) when:

$$\frac{2\mathcal{P}_{\text{beam}}}{c} = mg \Rightarrow \mathcal{P}_{\text{beam}} = \tfrac{1}{2}mgc$$

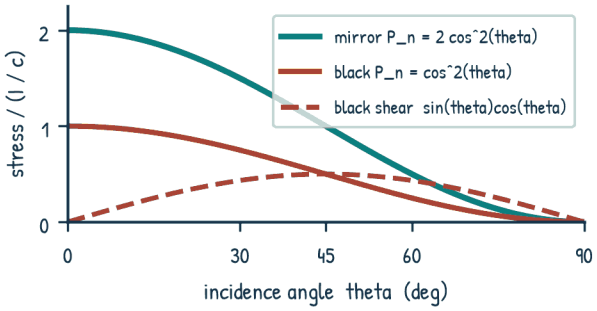
WHY / Because c = 3e8 m/s is huge, levitating just m = 1 mg requires P = 0.5 * 1e-6 * 9.8 * 3e8 = 1.47 kW of continuous reflected optical power!

WATCH / If a black plate heats up and re-radiates equally from BOTH faces, thermal emission recoils cancel; if insulated on one back face, isotropic emission adds recoil!

Two cosines on a flat sail: area projection cos(theta) times normal transfer



Normal pressure and tangential shear vs incidence angle theta from normal



REMEMBER / Actual flat area A_s at angle theta feels normal pressure P_n = (1+R)(I/c)cos^2(theta) and shear (1-R)(I/c)sin(theta)cos(theta); if P_hit is given, drop ONE cosine!

Electromagnetic spectrum, sources & applications

Adopted classroom wavelength cutoffs, physical emission mechanisms, RADAR ranging, and origin-overlap traps.

Seven spectral bands & six memorisable cutoffs

In decreasing vacuum wavelength lambda_0 (increasing frequency f = c/lambda_0 and photon energy E_gamma = h f = 1240 eV nm / lambda_0):

λ_0 cuts: $1\,\mathrm{m},\ 1\,\mathrm{mm},\ 700\,\mathrm{nm},\ 400\,\mathrm{nm},\ 10\,\mathrm{nm},\ 0.01\,\mathrm{nm}$

WHY / Corresponding frequencies are 300 MHz, 300 GHz, 429 THz, 750 THz, 30 PHz, and 30 EHz. Quick microwave rule: f(GHz) * lambda_0(cm) = 30 (2.4 GHz -> 12.5 cm, Q7)!

WATCH / Band boundaries are human conventions, NOT sudden changes in Maxwell laws; all seven travel at identical speed c = 299,792,458 m/s in vacuum (Q8).

Production mechanisms (NCERT + JEE Main/Adv core)

Match every band to its primary physical generator: Radio = accelerated charges in conducting antennas / LC circuits; Microwave = klystron, magnetron, Gunn diode.

Infrared = hot bodies & molecular vibrational-rotational transitions; Visible = outer valence electronic jumps; UV = inner valence jumps, mercury lamps, excimer lasers (193 nm LASIK, Q15).

$\mathrm{X-ray:}\ e^-\ \mathrm{decel\ /\ inner\ shells};\quad \gamma\mathrm{-ray:}\ \mathrm{nuclei\ /}\ e^+e^-$

WATCH / Origin test (E10): a 100 keV photon (lambda_0 = 0.0124 nm) emitted by a decelerating electron is an X-ray; the exact same 100 keV photon from a nucleus is a gamma ray!

RADAR echo ranging, greenhouse & microwave heating

A pulse emitted at t = 0 reflects off a target at range r and returns after echo delay dt: total path traveled by the wave is 2 r, NOT r (E9, Q27, Q35):

$$r_{\mathrm{radar}} = \frac{c\,\Delta t}{2}, \qquad E_\gamma(\mathrm{eV}) = \frac{1240}{\lambda_0(\mathrm{nm})}$$

Worked numbers: dt = 8.00 us gives r = 1200 m (Q27); KrF UV laser at 248 nm has E_gamma = 1240/248 = 5.00 eV (Q28); Cu-target X-ray at 0.100 nm has 12.4 keV (Q35).

WHY / Microwave ovens drive polar water molecules via oscillating E torques (dielectric loss, NOT simple sharp gas-phase resonance); greenhouse gases absorb/re-emit thermal IR (~10 um).

Atmospheric windows, ozone & ionospheric reflection

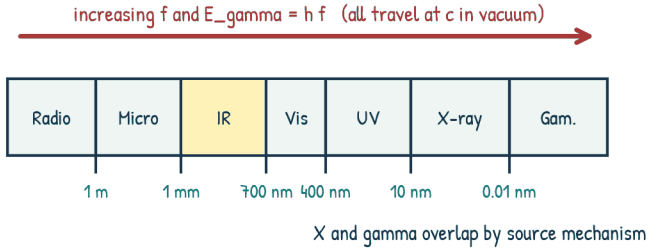
Stratospheric ozone O3 absorbs solar UV-B/C (~200-310 nm), protecting life; ordinary glass transmits visible light but blocks most thermal IR and short UV.

$$f_{\mathrm{crit,\,iono}} \approx 9\sqrt{N_{\max}\,(\mathrm{m^{-3}})}\ \mathrm{Hz}$$

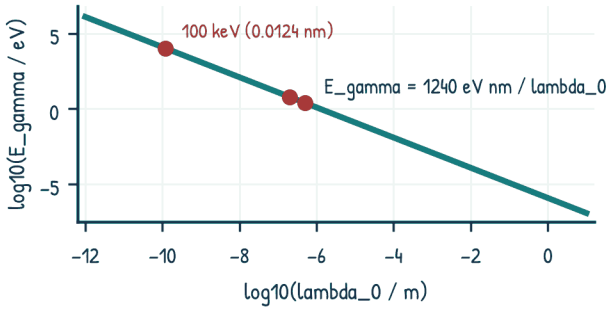
WHY / Shortwave HF skywaves (~3-30 MHz) undergo total internal reflection off ionospheric free electrons, whereas VHF/TV/microwaves (> 40 MHz) pierce straight through to satellites!

WATCH / Do not claim "thermal bodies emit ONLY infrared": hot filaments and the 5800 K Sun emit thermal radiation peaking in the visible while spanning IR through UV!

Adopted classroom wavelength cutoffs along the continuous vacuum EM spectrum



Exact inverse scaling E_gamma = hc / lambda_0 across thirteen wavelength decades



REMEMBER / All bands obey c = f lambda_0 in vacuum; X-ray vs gamma-ray naming follows physical origin (electronic vs nuclear), so their energies overlap.

LC oscillators, accelerating charges & dipole antennas

Why acceleration is required for radiation, Hertzian dipole angular pattern sin^2(theta), and w^4 Rayleigh/antenna scaling.

Lumped LC tank vs open radiating dipole

In an ideal closed LC circuit, Kirchhoff loop balance L d^2q/dt^2 + q/C = 0 sets the oscillation frequency of the carrier wave:

$$\omega_0 = \frac{1}{\sqrt{LC}}, \qquad f_0 = \frac{1}{2\pi\sqrt{LC}}, \qquad \lambda_0 = 2\pi c\sqrt{LC}$$

WHY / Opening the capacitor plates out into a straight half-wave rod of length L_ant ~ lambda_0 / 2 lets oscillating fields detach into space efficiently instead of staying trapped.

WATCH / A sub-wavelength loop/rod of size d << lambda_0 is a terrible radiator unless matched: efficient dipole antennas use L_ant = lambda_0 / 2 or quarter-wave monopole lambda_0 / 4!

Why ONLY accelerated charges radiate energy

Integrate Poynting flux over a sphere of huge radius r -> infinity: surface area grows as 4 pi r^2, so nonzero total power P_rad requires S ~ 1/r^2 and hence |E|, |B| ~ 1/r:

$$E_{\text{stat}} \propto r^{-2} \Rightarrow S \propto r^{-4} \to 0; \qquad E_{\text{rad}} \propto \frac{qa_\perp}{r} \Rightarrow P_{\text{rad}} > 0$$

WHY / A charge moving at constant velocity v carries its convective field along with it without shedding energy (verified immediately by jumping to its rest frame!).

Larmor power formula for a single non-relativistic point charge q with acceleration a: P_rad = mu0 q^2 a^2 / (6 pi c).

Hertzian oscillating electric dipole pattern

For a short dipole p(t) = p0 cos(wt) z_hat along the z-axis (size d << lambda << r), far-field intensity depends on polar angle theta from z_hat and frequency w:

$$I(r,\theta) = \frac{\mu_0 p_0^2 \omega^4}{32\pi^2 c}\frac{\sin^2\theta}{r^2}, \qquad \mathcal{P}_{\text{tot}} = \frac{\mu_0 p_0^2 \omega^4}{12\pi c}$$

WHY / Along the dipole axis (theta = 0 or pi), accelerated charges move purely radially toward the observer, so transverse projection a_perp = a sin(theta) vanishes: ZERO axial radiation!

Polarisation check: far-field E lies along polar unit vector theta_hat (in the plane containing z_hat and r_hat), while B wraps azimuthally along phi_hat.

Blue sky & red sunset: the w^4 Rayleigh bridge

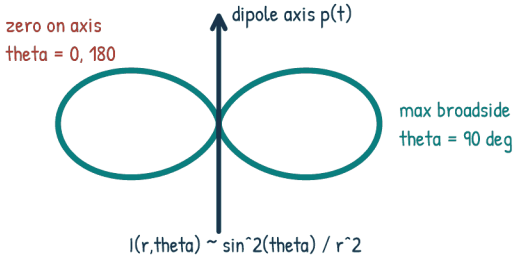
Sunlight drives bound electrons in atmospheric molecules as tiny induced dipoles p0 = alpha E_inc. Since P_scat ~ w^4 ~ lambda_0^-4:

$$\frac{I_{\text{violet (400 nm)}}}{I_{\text{red (700 nm)}}} \approx \left(\frac{700}{400}\right)^4 \approx 9.4$$

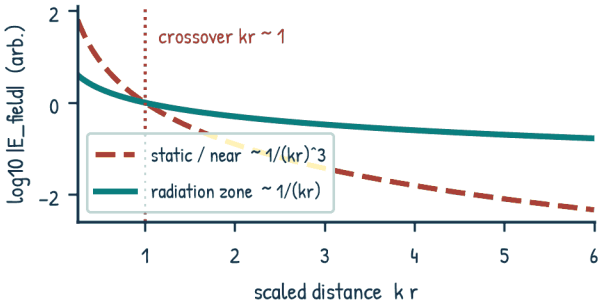
WHY / Short blue/violet wavelengths scatter ~6-9x more strongly sidewards (blue sky), leaving direct long-path sunset beams enriched in unscattered red/orange light.

WATCH / Notice directional ratio for dipole vs isotropic: since <sin^2(theta)>_sphere = 2/3, peak equatorial intensity I_max(theta = 90 deg) = (3/2) * P_tot / (4 pi r^2)!

Hertzian electric dipole pattern: maximum broadside, zero along the oscillation axis



Near-field terms die as 1/r^3; only the 1/r acceleration field escapes to infinity



REMEMBER / Stationary charges give E ~ 1/r^2, steady currents give B ~ 1/r^2; ONLY accelerating charges produce 1/r fields that carry power to infinity.

EM waves master triage, scales & examiner traps

Complete 6-branch problem playbook, mental-math constants, and the ten highest-frequency JEE Advanced traps.

Fast mental-arithmetic constants for exams

Keep six pre-combined combinations ready so you never multiply raw powers of 8.85e-12 and 4 pi e-7 under time pressure:

$$Z_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} \approx 377\ \Omega \approx 120\pi\ \Omega, \qquad \frac{1}{\varepsilon_0 c} = Z_0$$

$$\frac{1\,\mathrm{W}}{c} = 3.33\,\mathrm{nN}, \qquad \frac{1\,\mathrm{kW/m^2}}{c} = 3.33\,\mu\mathrm{Pa}, \qquad hc \approx 1240\,\mathrm{eV\,nm}$$

WHY / Look how instant E5 becomes with Z0 = 120 pi: for E0 = 120 V/m, I = 120^2 / (2 * 120 pi) = 60 / pi = 19.10 W/m^2 in three seconds flat!

Top amplitude & wave-equation traps

WATCH / Peak vs RMS factor of 2: I = E0^2 / (2 Z) = E_rms^2 / Z, and <u> = (1/2) eps E0^2 = eps E_rms^2. Confusing the two drops half the marks on numericals.

WATCH / Medium ratio E0 = v B0: in water/glass of index n, B0 = n E0 / c, NOT E0 / c!

WATCH / Divergence test: F(kx - wt) solves the 1D wave PDE, yet E = F(kx - wt) x_hat is NOT an EM wave in free space because k . E != 0!

WATCH / Coherent superposition: add vector fields E_tot = E1 + E2 BEFORE squaring; never add intensities directly when two coherent beams interfere or form a standing wave!

Top force, cosine & surface traps

WATCH / Cosine count: given intensity I on tilted area A_s -> F_n = (1+R)(I A_s/c)cos^2(theta). Given total beam power P_hit landing on tilted mirror -> F_n = 2(P_hit/c)cos(theta)!

WATCH / Absorber direction: a tilted black plate feels force straight along incident beam k_hat (both normal P_n and shear F_t != 0), NOT perpendicular to the plate!

WATCH / Polished sphere vs flat mirror: both black sphere AND specular sphere feel F = I pi a^2 / c; only a flat normal mirror feels 2 I pi a^2 / c!

WATCH / Transparent plate: use P_n = (1 + R - T)(I/c)cos^2(theta), NEVER (1 + R)I/c when transmitted fraction T > 0 carries forward momentum away!

Five-step pre-submission check protocol

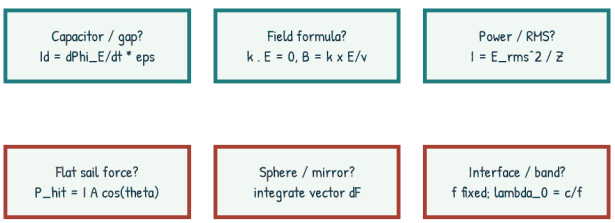
1. Spanning surface check: did you evaluate Ic and Id on ONE surface bounded by C, respecting fixed-I vs fixed-dV/dt constraints?

2. Triad check: did you test k . E = 0 and verify E_hat x B_hat points along the sign of constant phase (+k_hat)?

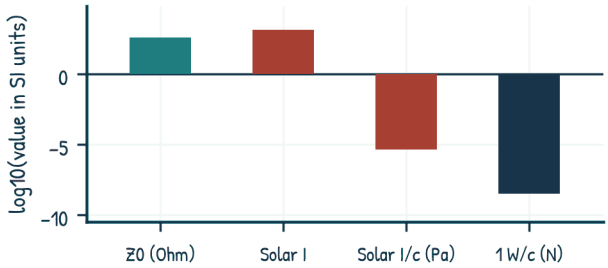
$$[I] = \mathrm{W/m^2}, \quad [u] = [P_{\mathrm{rad}}] = \mathrm{J/m^3} = \mathrm{Pa}, \quad [p] = \mathrm{N \cdot s}$$

3. Limiting check: does P_n vanish as theta -> 90 deg, recover I/c at R = 0, and double to 2I/c at R = 1?

Six-way JEE triage: identify the model before writing any equation



Order-of-magnitude anchors: Z0 ~ 377 Ohm, sunlight ~ 1.4 kW/m^2, 1 W/c ~ 3.3 nN



REMEMBER / Before substituting numbers, write four tags on paper: [medium n, Z], [PEAK vs RMS], [k_hat sign], and [beam-normal A vs tilted A_s].

Radial Poynting flux in capacitors, wires & coax

How circuits truly transport energy through empty space around wires, exact Q29 gap flux proof, and coaxial TEM guides.

Charging capacitor: inward radial energy flux (Q29)

Take circular vacuum plates of radius a, spacing d << a, E = E z_hat and azimuthal B_phi(a) = mu0 Ic / (2 pi a) phi_hat at cylindrical rim r = a:

$$\mathbf{S}(a) = \frac{E\hat{\mathbf{z}} \times B_\phi(a)\hat{\boldsymbol{\phi}}}{\mu_0} = -\frac{EI_c}{2\pi a}\hat{\mathbf{r}}$$

$$\mathcal{P}_{\text{in}} = |\mathbf{S}(a)|\,(2\pi a d) = (Ed)\,I_c = VI_c = \frac{d}{dt}\left(\frac{q^2}{2C}\right)$$

WHY / Notice z_hat x phi_hat = -r_hat! Electrostatic energy stored in the gap enters sideways from the surrounding field across the rim area 2 pi a d.

Cylindrical Ohmic wire: radial inflow feeds Joule heat

A steady DC current I runs along +z through a cylindrical wire of radius a, length l, resistance R = V/I: inside E = (V/l) z_hat and surface B_phi(a) = mu0 I/(2 pi a):

$$\mathbf{S}(a) = -\frac{VI}{2\pi a\ell}\hat{\mathbf{r}} \Rightarrow \mathcal{P}_{\text{in}} = |\mathbf{S}(a)|(2\pi a\ell) = VI = I^2R$$

WHY / Battery fields outside the metal funnel energy radially inward through the wire surface at rate I^2 R, where collisions convert it into lattice heat!

Inside radius r < a, E(r) = V/l is uniform while B_phi(r) ~ r, so inward power across radius r is V I (r^2/a^2) — proportional to enclosed volume!

Coaxial cable: lossless TEM wave power transport

Inner core radius a (+I along +z, potential +V) and outer shield radius b (-I, potential 0) carry radial E_r and azimuthal B_phi in insulator (eps, mu):

$$E_r = \frac{V}{r\ln(b/a)}, \qquad B_\phi = \frac{\mu I}{2\pi r}, \qquad S_z = \frac{VI}{2\pi r^2\ln(b/a)}$$

$$\int_a^b S_z\,(2\pi r\,dr) = VI, \qquad Z_{\text{coax}} = \frac{V}{I} = \frac{1}{2\pi}\sqrt{\frac{\mu}{\varepsilon}}\ln\left(\frac{b}{a}\right)$$

WHY / Integrating S_z across the annular dielectric gives exact circuit power V I! For vacuum coax, Z_coax = 60 ln(b/a) Ohm (50 Ohm standard at b/a = 2.30).

Hidden magnetic-energy term & validity fence

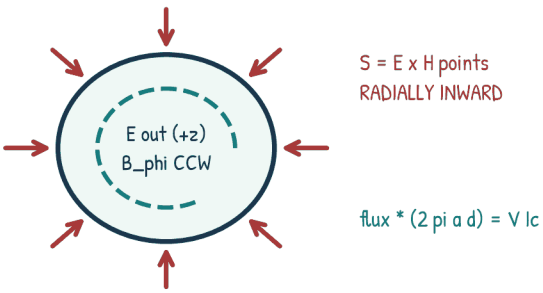
In Q29 with non-steady current q(t) = alpha t^2 (Ic = 2 alpha t), changing B_phi inside the gap induces a small second-order radial correction to E(r):

$$U_B = \int_0^a \frac{B_\phi^2}{2\mu_0}\,2\pi r d\,dr = \frac{\mu_0 I_c^2 d}{16\pi}, \qquad \frac{U_B}{U_E} \sim \left(\frac{\omega a}{c}\right)^2$$

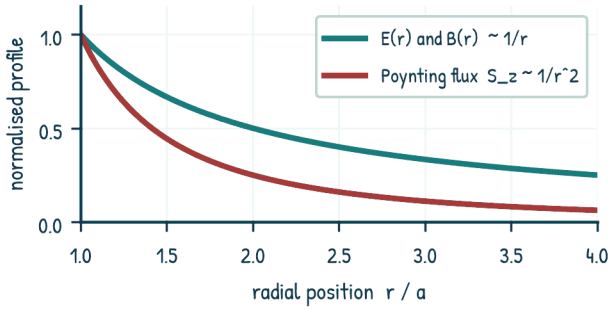
WHY / When variation time tau >> a/c (w a << c), U_B << U_E justifies Model Q. Retention of dU_B/dt gives the next term in the cylindrical cavity expansion!

WATCH / Never extrapolate quasistatic gap fields to t -> 0 shock steps where dI/dt is impulsive: sharp wavefronts travel radially inward at speed c!

Charging capacitor top view: field energy enters radially through the cylindrical rim



Coaxial TEM line: fields fall as 1/r between conductors a and b, carrying total V I



REMEMBER / Wires guide fields, while energy travels in the surrounding space via S = E x H: inward into capacitors/resistors and axially along coax.

Curved-surface forces, cones & stellar beta blow-out

Exact integrals proving F_black = F_specular = I pi a^2/c on a sphere, cone/cylinder drag factors, and astrophysical radiation blow-out.

Why black & specular spheres feel identical axial force

Divide the sunlit hemisphere of radius a >> lambda into ring zones dA = 2 pi a^2 sin(theta) d(theta) at normal angle theta in [0, pi/2] (sec. 2.11, Q14):

$$F_{\text{abs}} = \frac{I}{c}\int_0^{\pi/2} \cos\theta\,(2\pi a^2 \sin\theta\,d\theta) = \frac{I\pi a^2}{c}$$

$$F_{\text{spec}} = \int_0^{\pi/2} \left(\frac{2I}{c}\cos^2\theta\right)\cos\theta\,(2\pi a^2 \sin\theta\,d\theta) = \frac{4\pi a^2 I}{c}\left[\frac{1}{4}\right] = \frac{I\pi a^2}{c}$$

WHY / Physical miracle: a ray hitting at theta scatters at angle pi - 2 theta from forward, carrying away axial momentum -(dU/c)cos(2 theta) whose hemisphere average vanishes!

Cones, cylinders & retroreflectors compared

Apply the exact same local vector projection dF_z = dF_n cos(theta) + dF_t sin(theta) across other symmetric bodies of projected cross-section A_proj:

$$F_{z,\,\text{cone}} = \frac{I\pi a^2}{c}(1 + R\cos 2\alpha_0) = \frac{I\pi a^2}{c}(1 - R + 2R\sin^2\alpha_0)$$

Here alpha_0 is the cone semi-vertical angle (normal makes theta = pi/2 - alpha_0 with axis). Check limits: alpha_0 = pi/2 (flat disc) -> (1 + R) I pi a^2 / c; sharp needle alpha_0 -> 0 with R = 1 -> 0!

WHY / Specular circular cylinder of radius a, length L perpendicular to beam: int_{-pi/2}^{pi/2} 2 cos^3(theta) d(theta) = 8/3 -> F = (4/3)(2 I a L / c) vs black 2 I a L / c!

Stellar radiation blow-out ratio beta (Q34)

Around a star of luminosity L_* and mass M_*, compare outward radiation force I pi a^2/c on a spherical dust grain (density rho_m, radius a) to inward gravity G M_* m / r^2:

$$\beta \equiv \frac{F_{\text{rad}}}{F_{\text{grav}}} = \frac{3L_\star}{16\pi c G M_\star \rho_m a}, \qquad a_{\text{crit}}(\beta = 1) = \frac{3L_\star}{16\pi c G M_\star \rho_m}$$

WHY / Both I = L_*/(4 pi r^2) and gravity fall strictly as 1/r^2, so orbital distance r CANCELS COMPLETELY! For Sun (L_sun = 3.83e26 W, M_sun = 1.99e30 kg) & silicates, a_crit ~ 0.2 um.

WATCH / Do not let beta -> infinity as a -> 0: once grain radius a ~ lambda (~0.5 um), geometric optics breaks down and Mie/Rayleigh scattering cross-section drops as a^6!

Lambertian diffuse reflector: third classic case

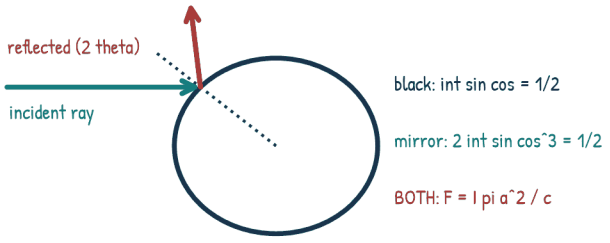
If an opaque flat disc (projected normal incidence) re-scatters all incident energy diffusely with Lambert cosine radiance (~ cos psi into hemisphere), outgoing normal momentum adds 2/3:

$$P_{\text{Lambert, normal}} = \left(1 + \tfrac{2}{3}R_{\text{diff}}\right)\frac{I}{c}$$

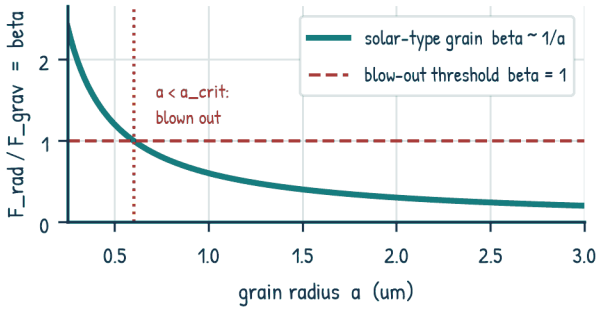
WHY / Diffuse bounce always sits strictly between total absorption (1 * I/c) and normal specular mirror (2 * I/c) because oblique scattered rays waste sideways momentum.

Exemplar check (E8): a = 1.00 mm grain, rho_m = 3000 kg/m^3 at I = 1200 W/m^2 gives F = 12.6 pN and a = 3 I/(4 rho_m a c) = 1.00e-6 m/s^2.

Illuminated sphere zone: local specular scatter equals full black-sphere drag



Radiation-to-gravity ratio beta cancels distance r^2 and scales inversely with radius a



REMEMBER / A specular sphere scatters light into all directions with <cos(2 theta)> = 0, feeling the EXACT same force I pi a^2 / c as a black sphere!

Wave impedance, Fresnel matching & quarter-wave coats

Boundary loops for tangential E and H, why t_E > 1 never violates energy conservation, and impedance-matching layers.

Tangential boundary equations & reflected H sign

Shrink rectangular Faraday and Ampere loops across a charge/current-free planar interface z = 0 between lossless media Z1 (z < 0) and Z2 (z > 0):

$$E_i + E_r = E_t, \qquad \frac{E_i - E_r}{Z_1} = \frac{E_t}{Z_2}$$

WHY / Keeping +x_hat for E_i, E_r, E_t means forward wave has H_i = +E_i/Z1 y_hat, while reflected wave moving along -z_hat has H_r = -E_r/Z1 y_hat!

$$r_E = \frac{Z_2 - Z_1}{Z_2 + Z_1} = \frac{n_1 - n_2}{n_1 + n_2} \ (\mu_1 = \mu_2), \qquad t_E = \frac{2Z_2}{Z_1 + Z_2}$$

Energy conservation & the "t_E > 1" paradox (Q16, Q36)

Intensity is I = E0^2 / (2 Z), so transmitted intensity carries factor 1/Z2 while incident carries 1/Z1. Dividing I_r and I_t by I_i proves exact flux balance:

$$R = r_E^2 = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1}\right)^2, \qquad T = \frac{Z_1}{Z_2} t_E^2 = \frac{4Z_1 Z_2}{(Z_1 + Z_2)^2}, \qquad R + T = 1$$

Paper Q36: for Z2 = 2 Z1, r_E = 1/3 and t_E = 4/3 > 1! Yet R = 1/9 and T = (1/2)(16/9) = 8/9 -> R + T = 1! Larger E amplitude inside larger Z carries LESS current H!

WATCH / Writing T = t_E^2 instead of (Z1/Z2) t_E^2 gives T = 16/9 > 1 — an instant violation of energy conservation whenever Z1 != Z2!

Equal speed != zero reflection in magnetic matter (Q36c)

Can a boundary reflect waves even when BOTH media have the exact same wave speed v1 = v2 (same refractive index n1 = n2)? YES, if permeabilities differ!

$$\mu_2 = 2\mu_1, \ \varepsilon_2 = \varepsilon_1/2 \Rightarrow v_2 = v_1, \quad Z_2 = 2Z_1 \Rightarrow R = 1/9$$

WHY / Conversely, choosing mu2/mu1 = eps2/eps1 makes Z2 = Z1 so R = 0 (invisible Stealth/impedance-matched boundary) EVEN THOUGH wave speed drops across the boundary!

Nonmagnetic glass n = 1.5 in air (Z2 = Z0/1.5): r_E = -0.20 (pi phase flip on E!), R = 0.04 (4% reflection per face), T = 0.96.

Quarter-wave anti-reflection transformer

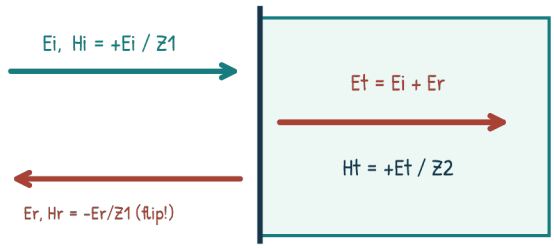
To eliminate reflection completely between media Z1 and Z3 at target vacuum wavelength lambda_0, insert a lossless layer (index n_c, impedance Z_c) of thickness d:

$$d = \frac{\lambda_{coat}}{4} = \frac{\lambda_0}{4n_c}, \qquad Z_c = \sqrt{Z_1 Z_3} \quad (n_c = \sqrt{n_1 n_3} \ \text{ if } \mu_r = 1)$$

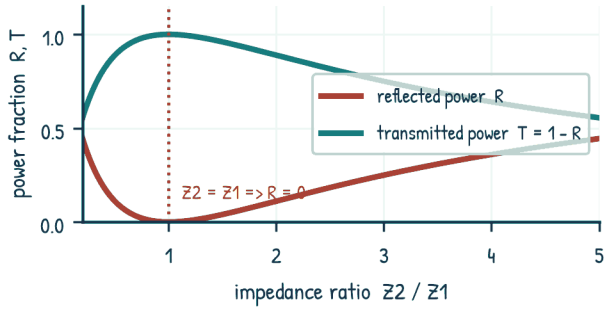
WHY / Equal reflection amplitudes at front and back faces suffer round-trip phase shift 2 k_c d = pi, cancelling destructively in reflection so 100% power transmits!

WATCH / If n1 < n_c < n3, BOTH front and back reflections carry the SAME pi phase flip, so destructive interference still requires 2 n_c d = lambda_0 / 2 (NOT lambda_0)!

Normal-incidence matching: tangential E and H stay continuous; reflected H flips sign



Power reflection R and transmission T vs impedance ratio Z2/Z1 (always R + T = 1)



REMEMBER / Reflection is governed by wave impedance Z = sqrt(mu/eps), NOT refractive index n alone; always compute power transmission as T = (Z1/Z2) t_E^2.

Standing EM waves, mirror stress & cavity pressure u/3

Constructing exact node structure in front of a conductor, magnetic Maxwell stress at a mirror, and 3D thermal photon gas.

Standing wave in front of a perfect mirror (z <= 0)

Superpose E_i = E0 cos(kz - wt) x_hat (+z) and reflected E_r = -E0 cos(kz + wt) x_hat (-z) so tangential E vanishes on mirror z = 0 (Q31):

$$\mathbf{E} = 2E_0 \sin(kz)\sin(\omega t)\,\hat{\mathbf{x}}, \qquad \mathbf{B} = \tfrac{2E_0}{c}\cos(kz)\cos(\omega t)\,\hat{\mathbf{y}}$$

WHY / For E_r moving along -z with -E0 x_hat, (-z_hat) x (-x_hat) = +y_hat: reflected B ADDS at z = 0 instead of cancelling!

Node map: E nodes at z = -m lambda/2 (including mirror z = 0); B nodes at z = -(2m + 1) lambda/4 (quarter-wave shifted!).

Zero mean Poynting vector vs nonzero mirror pressure

Multiply E and B: S_z = (2 E0^2 / Z0) sin(2kz) sin(2wt) sloshes energy back and forth every quarter-cycle with <S_z> = 0, while time-averaged density is spatially uniform:

$$\langle u \rangle = \varepsilon_0 E_0^2 = 2\langle u_{\mathrm{inc}} \rangle, \qquad P_{\mathrm{mirror}} = \left\langle \tfrac{B^2(0,t)}{2\mu_0} \right\rangle = \varepsilon_0 E_0^2 = \tfrac{2I_{\mathrm{inc}}}{c}$$

WHY / At z = 0, E = 0 so surface currents feel Lorentz force (J_s x B_avg_across_sheet) equal to magnetic Maxwell stress B(0,t)^2 / (2 mu0), matching momentum reversal 2 I_inc / c!

WATCH / Never write uE = uB in a standing wave: at wt = 0 ALL energy is magnetic; at wt = pi/2 ALL energy is electric!

Isotropic cavity radiation: why pressure is u / 3

Fill an enclosure uniformly with isotropic radiation of total energy density u: solid angle dOmega carries density du = u dOmega/(4 pi) at angle theta to a specular wall:

$$P_{\mathrm{iso}} = \int_{\theta<\pi/2} \tfrac{2\,du}{1}\cos^2\theta = \tfrac{2u}{4\pi}\int_0^{2\pi} d\phi \int_0^{\pi/2} \cos^2\theta \sin\theta\, d\theta = \tfrac{u}{3}$$

WHY / Only half the rays move toward the wall (factor 1/2), and they strike at all angles with <cos^2(theta)> = 1/3: (1/2) * 2 * (1/3) u = u / 3 (vs u for a collimated beam)!

WATCH / Careful boundary check (sec. 4.3): a cold black wall absorbing an isotropic field of total density u feels u / 6 until thermal equilibrium emission adds the second u / 6!

Adiabatic photon-gas expansion & cavity resonators

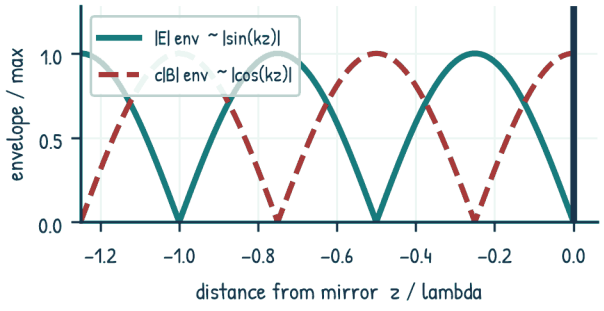
Combining P = u/3 and U = u V inside reversible First Law dU + P dV = T dS = 0 (with u = a T^4) gives the relativistic photon-gas adiabat:

$$PV^{4/3} = \text{const}, \qquad TV^{1/3} = \text{const} \quad (\gamma_{\mathrm{photon}} = \tfrac{4}{3})$$

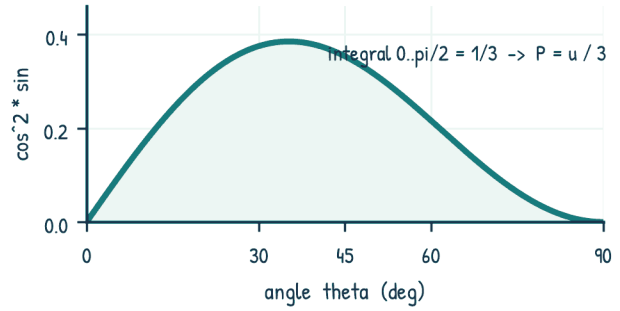
Resonant frequencies of a lossless rectangular metallic box (sides Lx, Ly, Lz):

$$f_{lmn} = \tfrac{c}{2}\sqrt{\left(\tfrac{l}{L_x}\right)^2 + \left(\tfrac{m}{L_y}\right)^2 + \left(\tfrac{n}{L_z}\right)^2}$$

Perfect mirror at z = 0 forces an E node and a B antinode (magnetic stress 2 I / c)



Angular weight cos^2(theta) sin(theta) peaks at 35.3 deg and integrates to 1/3



REMEMBER / In a standing EM wave E and B oscillate 90 deg out of phase in BOTH space and time: <S> = 0, yet mirror magnetic pressure equals 2 I_inc / c.

Conducting media, skin depth & collisionless plasmas

Damped telegrapher wave equation, 45 deg B-to-E phase lag in metals, shielding attenuation, and ionospheric plasma cutoff.

Wave equation in an Ohmic conductor (sec. 4.4)

Insert J_f = sigma E into Maxwell-Ampere inside a homogeneous conductor: curl B = mu sigma E + mu eps dE/dt. Taking curl of Faraday yields:

$$\nabla^2\mathbf{E} = \mu\sigma\frac{\partial\mathbf{E}}{\partial t} + \mu\varepsilon\frac{\partial^2\mathbf{E}}{\partial t^2}, \qquad k^2 = \mu\varepsilon\omega^2 + i\mu\sigma\omega$$

WHY / First time derivative breaks time-reversal symmetry (Ohmic dissipation!), making wavenumber k = k_R + i k_I complex so exp(i k z) = exp(i k_R z) exp(-k_I z) attenuates!

Poor-conductor limit (sigma << w eps, e.g. dry soil / seawater at high f): k_I ~ (sigma / 2) sqrt(mu/eps) = sigma Z / 2 is independent of frequency!

Good-conductor skin depth delta & 45 deg magnetic lag

When conduction current dominates displacement current (sigma >> w eps), k^2 ~ i mu sigma w. Since sqrt(i) = (1 + i)/sqrt(2), k_R = k_I = 1/delta:

$$\delta = \sqrt{\frac{2}{\mu\sigma\omega}}, \qquad Z_{\text{metal}} = \frac{\omega\mu}{k} = \sqrt{\frac{\omega\mu}{2\sigma}}(1-i) = \frac{1-i}{\sigma\delta}$$

WHY / Three huge Olympiad consequences: (1) E decays by 1/e every delta and intensity I decays as exp(-2 z/delta); (2) B lags E by pi/4 (45 deg!); (3) magnetic energy uB >> uE inside metal!

Scale check in copper (sigma ~ 5.8e7 S/m): delta ~ 8.5 mm at 60 Hz, ~ 66 um at 1 MHz, and ~ 2.1 um at 1 GHz!

Collisionless cold plasma dispersion & total reflection

In a neutralised gas of free electrons (number density n_e, mass m_e), Newton law m_e dv/dt = -e E gives J = -n_e e v = (i n_e e^2 / m_e w) E (purely inductive, ZERO heating!):

$$\varepsilon_{\text{eff}}(\omega) = \varepsilon_0\left(1 - \frac{\omega_p^2}{\omega^2}\right), \qquad \omega_p = \sqrt{\frac{n_e e^2}{m_e\varepsilon_0}}, \qquad \omega^2 = \omega_p^2 + c^2k^2$$

WHY / For w < w_p, eps_eff < 0 so k is purely imaginary: incident radio waves suffer 100% lossless reflection off the ionosphere or metals below UV plasma edge!

Phase speed > c vs group speed < c check

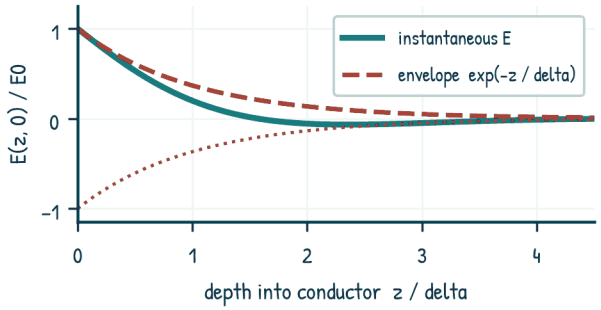
Above plasma frequency (w > w_p), differentiate w^2 = w_p^2 + c^2 k^2 with respect to k to compare crest velocity v_p = w/k with packet/energy velocity v_g = dw/dk:

$$v_p = \frac{c}{\sqrt{1-\omega_p^2/\omega^2}} > c, \qquad v_g = c\sqrt{1-\frac{\omega_p^2}{\omega^2}} < c, \qquad v_p v_g = c^2$$

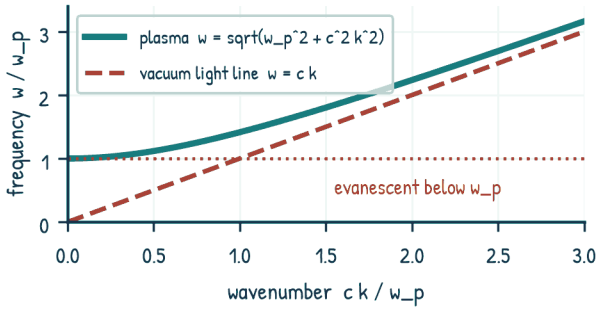
WHY / Relativity forbids signals/energy travelling faster than c — and group/signal velocity v_g stays strictly < c everywhere while crests slide faster along the envelope!

WATCH / In dispersion problems, never plug phase index n = c/v_p < 1 into packet travel delay: radio pulsar dispersion delay uses t_arrival = L / v_g!

Good conductor (k = (1+i)/delta): wave decays by 1/e in one skin depth delta



Collisionless plasma dispersion: v_p > c above cutoff w_p while group speed v_g = c^2/v_p < c



REMEMBER / Good conductors (sigma >> w eps) damp fields exponentially in skin depth delta = sqrt(2/(mu sigma w)); collisionless plasmas reflect below w_p with v_p v_g = c^2 above.

Relativistic Doppler, moving mirrors & waveguides

Exact Lorentz light shift, double boost on moving mirrors, Poynting-Robertson orbital spiral, and parallel-plate TE modes.

Relativistic optical Doppler & transverse time dilation

Because light requires no material medium, source-moving and observer-moving shifts are IDENTICAL and include Lorentz gamma = 1/sqrt(1 - beta^2), beta = u/c:

$$f_{\text{approach}} = f_0\sqrt{\frac{1+\beta}{1-\beta}}, \qquad f_{\text{recede}} = f_0\sqrt{\frac{1-\beta}{1+\beta}}, \qquad f_\perp = \frac{f_0}{\gamma}$$

WHY / Even at closest approach (90 deg in observer frame), clock time dilation slows source oscillations by 1/gamma = sqrt(1 - beta^2) — a pure second-order redshift!

Low-speed check (beta << 1): df / f0 = +- beta + O(beta^2); astronomical redshift parameter 1 + z = lambda_obs / lambda_emit = sqrt((1+beta)/(1-beta)).

Reflection & pressure on a moving flat mirror

A flat perfect mirror moves at speed u = beta c TOWARD an incoming normal vacuum beam of frequency f and intensity I (or minus sign if receding):

$$f' = f\frac{1+\beta}{1-\beta}, \qquad P_{\text{moving mirror}} = \frac{2I}{c}\frac{1+\beta}{1-\beta}$$

WHY / Two clean steps: mirror intercepts crests at relative speed c + u -> rate factor (1 + beta); reflected photon energy also jumps by (1 + beta)/(1 - beta), doing mechanical work on the beam!

Check beta -> -1 (mirror fleeing at light speed): f_ref -> 0 and P -> 0 because photons can never catch the mirror!

Poynting-Robertson drag: why dust spirals INTO stars

A black dust grain of radius a in circular orbit at speed v around a star absorbs radial photons in the star frame, but re-emits thermally isotropically in its OWN rest frame:

$$\mathbf{F}_{\text{PR}} \approx -\frac{I\pi a^2}{c}\frac{\mathbf{v}}{c} = -\frac{L_\star a^2}{4c^2 r^2}\mathbf{v}$$

WHY / In the star frame, forward-beamed thermal emission carries tangential momentum m_eff v = (P_abs/c^2) v away! Or equivalently in grain frame: stellar aberration tilts sunlight slightly head-on by angle v/c!

Result: angular momentum bleeds steadily so surviving micron grains (beta < 1/2) spiral inward over cosmic timescales.

Parallel-plate metallic waveguide (TE_m modes)

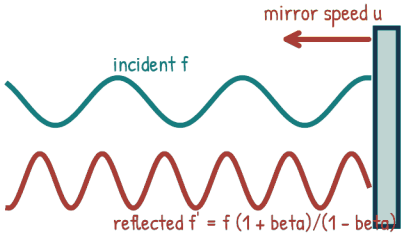
Confine waves between two perfectly conducting plates at y = 0 and y = b: vanishing tangential E_z/E_x on both walls quantises k_y = m pi / b (integer m >= 1 for TE):

$$k_z = \sqrt{\frac{\omega^2}{c^2} - \left(\frac{m\pi}{b}\right)^2}, \qquad \omega_{c,m} = \frac{m\pi c}{b}, \qquad \lambda_{c,m} = \frac{2b}{m}$$

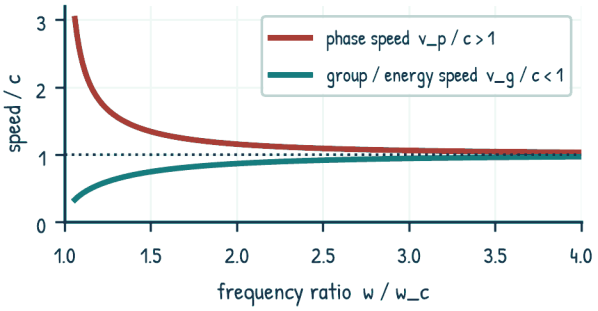
WHY / Picture a plane wave zig-zagging between walls at angle sin(theta_m) = w_c / w: crests slide faster along z (v_p = c / cos theta_m > c) while energy advances at v_g = c cos theta_m < c!

WATCH / A hollow single-conductor pipe has NO TEM mode (m = 0): if lambda_0 > 2 b for parallel plates, ALL TE modes are cutoff and decay exponentially!

Advancing mirror compresses reflected crests twice: double Doppler boost (1+beta)/(1-beta)



Guided TE_1 mode above cutoff w_c = pi c / b: exact product law v_p v_g = c^2



REMEMBER / In vacuum no air frame exists: Doppler depends ONLY on relative beta = u/c, and a moving mirror shifts both frequency and radiation pressure by (1+beta)/(1-beta).